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Range improvement in BEV trucks using heat pump for cabin heating instead of PTC heater

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ABSTRACT

The Paris Agreement (2016) and the European Green Deal (2018) have accelerated the development of zero-emission technologies in the transportation sector, particularly for heavy-duty vehicles. Battery electric vehicles (BEVs), while offering promising emission reductions, suffer from range limitations under cold ambient conditions due to the significant energy demand for cabin heating. This study investigates the performance of a heat pump system as an alternative to the conventional Positive Temperature Coefficient (PTC) heater in a heavy-duty electric truck application. A detailed one-dimensional (1D) vehicle thermal model is developed using GT-Suite to simulate the dynamic behavior of a 16-ton BEV truck equipped with a 300 kWh battery pack. The simulation is conducted under 0 °C ambient conditions using the On-Road Fuel Economy driving cycle. Compared to the PTC heater configuration, the heat pump system achieved a 31.64% reduction in auxiliary thermal energy consumption, resulting in a 1.18% improvement in overall battery energy consumption. These improvements translate into a measurable increase in driving range under cold climate scenarios. This study demonstrates the feasibility and effectiveness of integrating heat pump systems in heavy-duty electric trucks, which has not been extensively covered in previous literature focused primarily on passenger vehicles. While numerous studies have explored EV thermal management using various simulation platforms, including widely adopted tools like MATLAB/Simulink and Modelica, a comprehensive investigation specifically for heavy-duty BEV trucks, particularly focusing on the detailed 1D thermal modeling approach adopted here, remains less common.

1. Introduction

Excessive CO2 emissions threaten humanity. Thanks to official efforts such as the Paris Agreement and the Green Deal Process, significant measures are taken in all sectors, including transportation.

Current regulations force Original Equipment Manufacturers (OEMs) to reduce CO2 fleet average values by 43% in 2030 compared to the baseline of 2019. As a natural consequence of the strict regulations, OEMs are made to design their future portfolio with zero emission vehicle technologies such as hydrogen powertrains (such as fuel cell and hydrogen internal combustion engines) and battery electric vehicles (BEVs). However, these innovative technologies have their advantages not only for the future, but also for the disadvantages such as cost, complexity, durability, packaging, range, etc. OEMs are currently focusing on these problems to eliminate the risk of encountering possible failures in the serial production stage.

Abolishing range anxiety in BEV usage is a perplexing problem to solve. OEMs are investigating innovative, cheap, and lower complexity solutions. However, detecting a perfect trade-off between these performance metrics is really hard. Covering stringent emission regulations and eliminating the risk of paying significant penalties sometimes requires directing more attention to energy efficiency rather than other aspects such as cost, complexity, and packaging.

Heat pump technology lies at a very beneficial point in terms of energy efficiency improvement. A heat pump is a way to heat or cool any space by moving heat from one spot to another. Instead of making heat from scratch, it simply transfers it, which makes it a smarter choice in terms of energy use compared to regular heating and cooling systems. In essence, a heat pump extracts heat from a source (such as air, ground, or water) and transfers it to a different location. During winter, it can draw heat from the outside air and bring it indoors to warm up a space. Due to these superior capabilities, the use of heat pump in automotive applications is always under consideration.

Heat pumps are used in thermal management systems for electric vehicle cabins [1]. The operational range of these vehicles may diminish by roughly 10% in cooling mode, whereas it may decrease by approximately 48% in heating mode [2,3]. Through the introduction of Yang et al. [4] in 2022, heating systems for pure electric vehicles (EVs)

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Nomenclature

Symbols and Units

m	Vehicle mass (tons)
r	Tire radius (m)
f_R	Rolling resistance coefficient
C_W	Drag coefficient
A	Vehicle frontal area (m^2)
ρ	Air density (kg/m^3)
g	Gravitational acceleration (m/s^2)
α	Gradient angle (degrees)
$\dot{m}_{air}$	Air mass flow rate (kg/s)
$P_{compressor}$	Compressor power (kW)
h_1, h_2, h_3, h_4	Enthalpy at states 1–4 (kJ/kg)
s_1, s_2, s_3, s_4	Entropy at states 1–4 (kJ/kg K)
$\dot{Q}_{cabin}$	Heat delivered to cabin (kW)
$\dot{Q}_{evap}$	Heat absorbed in evaporator (kW)
A_{fr}	Heat exchanger frontal area (m^2)
H, W, D	Heat exchanger height, width, depth (mm)
I	Exergy destruction (kW)

Abbreviations

BEV	Battery Electric Vehicle
COP	Coefficient of Performance
EV	Electric Vehicle
GT	Gamma Technologies Suite
HT	High Temperature
HV	High Voltage
HVAC	Heating, Ventilation, and Air Conditioning
HP	Heat Pump
ITS	Integrated Thermal System
LT	Low Temperature
LV	Low Voltage
OCV	Open Circuit Voltage
PTC	Positive Temperature Coefficient
SOC	State of Charge
TMS	Thermal Management System
ORFE	On-Road Fuel Economy
EXV	Electronic Expansion Valve
TXV	Thermostatic Expansion Valve
VPI	Vapor Injection
WLTC	Worldwide harmonized Light vehicles Test Cycle
R134a, R744	Refrigerants (Tetrafluoroethane, CO_2)

were given a detailed review that covered the old and new technologies, improvements in current energy efficiency and extension of the vehicle range. The article analyzed those widely applied heaters known for their poor performance such as the positive temperature coefficient (PTC); these were ranked with state of the art heat pump systems that promised relatively enhanced efficiencies except for their low temperature performances, and also reviewed new heating developments such as adsorptive air conditioning, combustion heating by fuel, heat storage, recovery of waste heat, thermoelectric heating systems, and magnetocaloric heating. Some have great potential for energy savings, while others may have several future research topics such as system efficiency, complexity of integration, and material limitations.

In 2024, Kharabati and Saedodin [5] had a systematic review of EV battery thermal management approaches which conclusively discussed the need for keeping optimal battery temperature to ensure performance, life, and safety. Battery Thermal Management System approaches, such as the air cooling, liquid cooling, and phase change material (PCM) based cooling, heat pipe, thermoelectric cooling, and heat pump, have been put forward for study. Air cooling is cheap, yet inefficacious while liquid cooling provides enhanced performance but ultracomplex. Heat pump systems used extensively for heating and cooling the passenger cabin of an EV were appraised for eligibility in application for battery thermal management system. Advances energy saving still have constrained performances at low temperature. The hybrid cooling system that could be PCM with heat pipes or liquid cooling was a very promising solution to present better energy savings and efficiency. However, challenges such as weight and cost, as well as material long-term stability, pose important issues for future research and development.

In 2025, Li et al. [6] published an experimental study regarding the frost characteristics and performance impact of R744 (CO2) heat pump systems in EV under low ambient temperature conditions. The study found that frosting on outdoor heat exchangers significantly affects efficiency, with lower ambient temperature and higher compressor speed resulting in rapid frost formation. The research also presented the Frost Coverage Ratio (FCR) as an important parameter, proving that whenever FCR is higher than 40%, there is a reduction in suction pressure at the compressor, a drop in heating capacity by 26%–31% and a decrease in Coefficient of Performance (COP) by 21%–29%. In addition, the study compared R744 heat pump systems with R134a heat pump systems, revealing that R134a, with more severe frosting, had a greater adverse effect on compressor performance and capacity for heating. The findings bring out the reality of optimizing defrosting strategies and innovations in material to mitigate the efficiency loss of frosting in the heat pump systems of EV.

Zhang et al. [7] examined the relationship between frost and water behaviors and the impact of temperature and humidity on the capacity and effectiveness of frost and defrost cycles in electric car transcritical CO2 heat pump systems. The distribution of the frost layer on the heat exchanger's surface is uneven, which is closely linked to variations in the occupation of liquid-phase refrigerant, according to experimental research done by Li et al. [8,9] on the frost and defrosting processes of the external heat exchanger in the R134a heat pump air conditioning system of electric vehicles.

In 2025, Teng et al. [10] recommended a dual electronic expansion valve (EXV1 and EXV2) control strategy that optimizes the heating performance of CO2 heat pump systems in electric vehicles at low ambient temperatures. The results suggest that at 20 °C, a well-defined control strategy made improvements in heating capacity, COP, and gas cooler outlet air temperature, while saving energy consumption by 34.7% compared to that of R134a system. The optimized system also increased the range of the vehicle by 9.3% and 16.6% at 20 °C and 10 °C, respectively. The results indicate that CO2 heat pump system provides better efficiency and sustainability for EV thermal management.

In 2025, Song et al. [11] proposed a multi-component coupling model for dynamic and steady performance analysis of an integrated thermal management system (TMS) using a CO2 heat pump for electric vehicles at low temperatures. A cabin heating discharge pressure was shown to have a maximum value of 62 bar, while pre-warming the battery in series enhanced system performance under subcritical conditions. Additionally, the performance increased by discharging waste heat from the motor alone with an optimum discharge pressure of 48 bar, leading to an efficient heating management that covers the cabin, battery, and motor under continuous Worldwide harmonized Light vehicles Test Cycle (WLTC) at 15 °C.

In 2025, Ji et al. [12] conducted an experimental study on the performance of an integrated vapor injection (VPI) R1234yf heat pump system for electric vehicles (EVs) under cold conditions. The study featured a new modular design for VPI that reduces complex piping,

and weight on the system, and gives improved efficiency. The research shows that the new VPI module lowers intermediate pressure by 34.3%, while it has a reduced optimal refrigerant charge between 15.2 and 37.1%. Exergy analysis revealed that with the introduction of VPI, exergy efficiency increases by 5.3 to 12.5 percent and maintaining higher efficiency even at −5 °C. Apart from these, the study also compared various strategies for enhancing performance and demonstrated that achieving conditions of opening VPI increases heating capacity by 19.7% when the compressor is running at higher speeds, as against a speed increase of 25 to 30 percent, which improves heating capacity by only 5.5%. The findings speak positively about the merits of VPI application in EV heat pump systems, indicating that an optimized control strategy embracing both VPI operation and compressor speed control can culminate into maximizing efficiency while minimizing energy losses.

In 2025, Liu et al. [13] studied R454C as a stand-in for R134a in secondary-loop heat pump systems for mobile air conditioning, noting that it has increased efficiency and decreased environmental impact and found that Global Warming Potential (GWP) is 146. They found that R454C exceeded R290 in performance in terms of heating capacity, giving a 19.2% higher heating capacity at 20 °C and COP of 3.08 at 35 °C. There was, however, COP degression at both temperatures, which may be attributed to poor heat exchange in the WCC and compressor mismatch in design. The conclusions of the study pointed to R454C as a possible alternative for energy savings and sustainability in automotive air conditioning systems.

The effects of a gas–liquid separator on the performance of heat pump air conditioning systems in electric vehicles under low-temperature conditions were assessed by Li et al. [14]. Their results exhibited that, for those systems, placing the separator ahead of the outdoor evaporator regulated evaporating pressure by optimizing the gas-phase bypass valve set point between 20% and 30% to achieve the maximum performance. The modification introduced was 6.9% increase in heat capacity, 7.4% increase in COP, and outlet air temperature improvements that ranged between 3.6% and 11.2%. These results could suggest that the integration of a gas–liquid separator could improve the performance of the system and negate the adverse effects of nonuniform frost formation on the evaporator surface.

The issue of frost development in external heat exchangers for electric vehicles has been extensively studied by numerous academics [15].

In 2025, Hamid et al. [16] experimentally investigated brazed plate heat exchangers as desorbers in absorption–compression heat pump systems. They found that optimizing desorber design under varying mass fluxes significantly improved heat transfer performance, achieving up to 86% thermal efficiency and making the system suitable for industrial waste heat recovery.

In 2020, Ishaque et al. [17] numerically studied the impact of outdoor heat exchanger geometry on seasonal heat pump performance. Their results showed that cylindrical designs improved airflow uniformity and increased seasonal efficiency by up to 7.07%, highlighting the role of geometry in optimizing heat pump systems.

In this paper, evaluation of using heat pump and PTC heater in determining energy efficiency and driving range of an electric vehicle in different operating condition. Intending to show energy consumption and performance in driving range that the two heating technologies will have under different climates and on different driving scenarios, the study examines two heating technologies regarding climatic conditions and driving scenarios.

The Heating, Ventilation, and Air Conditioning (HVAC) system examined includes a reversible heat pump the efficiency of which is said to be optimized due to its very good thermal management capabilities and a PTC heater, which has been very effective in driving fast heat application and high energy consumption. Then the comparison is given over both methods, energy savings potentials, and range improvements to be expected.

Energy consumption and associated driving range reductions resulting from this will dynamically evaluate different conditions and settings of usage. These accordingly give a good guideline for optimizing the thermal management systems on electric vehicles with a consequence of overall efficiency improvement and enhanced driving range.

While the benefits of replacing PTC heaters with heat pump systems have been widely explored in passenger electric vehicles, limited research has addressed this integration in the context of heavy duty electric trucks. Due to their significantly different duty cycles, higher thermal inertia, larger battery capacities, and distinct operational constraints, the performance and efficiency impacts of cabin heating systems in Heavy Duty Trucks cannot be directly extrapolated from passenger vehicle studies.

Furthermore, existing literature has primarily concentrated on waste heat utilization or battery preconditioning in light-duty platforms, without a comprehensive evaluation of heat pump-based cabin heating for heavy-duty applications. In the design and optimization of thermal management systems, simulation environments like MATLAB/Simulink and Modelica are widely used. For example, Rahmani et al. [18] investigated battery thermal management, and Kiss et al. [19] modeled general EV thermal systems, both utilizing MATLAB/Simulink. However, existing literature primarily focuses on passenger vehicles or battery-specific thermal management. Comprehensive one dimensional (1D) thermal modeling for challenging applications like cabin heating in heavy duty BEV trucks remains limited. Therefore, this study aims to bridge this gap by developing a detailed 1D thermal model for a BEV truck and assessing the performance improvement achieved through heat pump implementation under realistic driving and environmental conditions.

However, the thermal management requirements for heavy duty BEV trucks differ significantly from those of passenger BEVs. Heavy-duty trucks feature substantially larger cabin volumes and operate for longer, often continuous, periods. This dramatically increases the absolute energy required for cabin conditioning in heavy-duty trucks. While passenger cars are often driven on short trips that can generate transient heating peaks, the long continuous operation and larger cabin volumes of trucks result in higher cumulative energy demand, thereby magnifying the impact of cabin conditioning on the total energy consumption compared to passenger cars. While studies on heat pump applications in passenger vehicles are available, the unique thermal load profiles and operational distinctions of trucks necessitate a dedicated investigation into the feasibility and efficiency potential of heat pump technology in this specific segment. Therefore, this study focuses explicitly on heavy duty BEV truck applications to evaluate the effects of a heat pump on energy efficiency and driving range under these demanding conditions, which are distinct from those of passenger cars.

2. One dimensional model of E-truck

Mahmud et al. [20] undertook the tests of different kinds of software packages designed to cater for different kinds of electrification studies, aggregating 125 different software packages for the simulation of electrified powertrains. The result of their analysis did indicate that one-dimensional tools such as GT-Suite are very effective when analyzing hybrid powertrains and internal combustion engines.

GT-Suite enabled the development of the e-truck model in this study, [21] whose extensive simulation capabilities coupled to a friendly user interface convinced us that this platform would be ideal for such task. Unlike general-purpose simulation environments such as MATLAB/Simulink, which often require extensive custom coding for complex physical phenomena, or Modelica, which excels in object-oriented modeling but may necessitate significant effort for detailed component libraries in automotive thermal management, it offers a specialized, pre-validated library of automotive components and thermal systems. This allowed for the rapid development of a high-fidelity 1D thermal model, enabling detailed dynamic behavior analysis of the

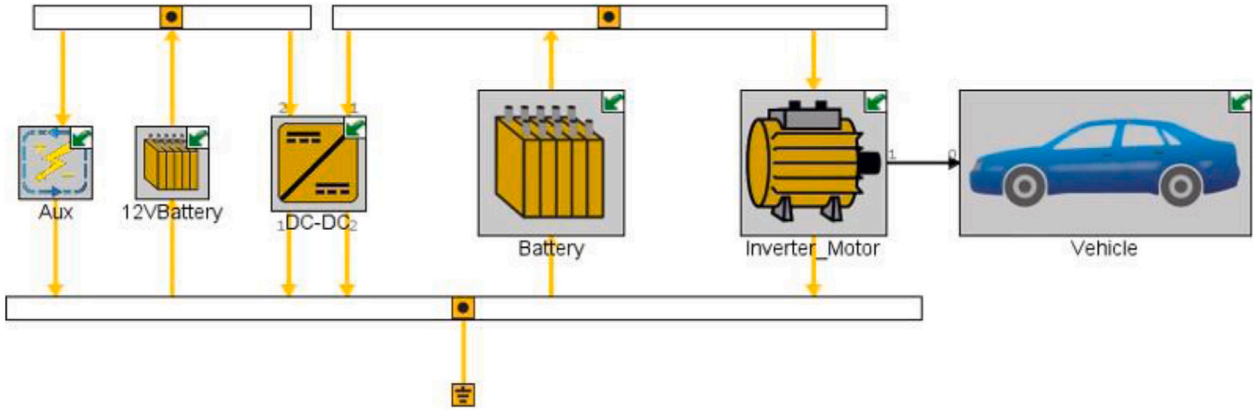

Fig. 1. E-truck 1D model.

entire vehicle system, from powertrain to HVAC, with a focus on real world operating conditions and complex interactions that might be more cumbersome to implement and validate in other environments. The thermodynamic model [22] is computer aided engineering software which offers a complete modeling environment for analyzing vehicle system performance.

The one-dimensional truck model calculates the necessary tractive wheel torque for a given target velocity and uphill gradient based on transient cycle data (with inputs such as speed–time and distance-slope). Depending on system efficiency and gear ratios, this force is transmitted to the electric engine as speed–torque. The efficiency of a motor is established with the help of an efficiency map, which would correlate with the applied torque and speed seconds. The required motor output power for this process will be coupled and drawn from the battery, which is similar to an open-circuit voltage (OCV), internal resistance maps, and real-time consumption calculation [23].

Important parts of the Electric Powertrain were modeled using thermodynamic model and supplied data. A battery capacity of 300 kWh was assigned. Fig. 1 depicts the one-dimensional (1D) e-truck model built in this study—as having an e-drive system with a 3-speed transmission. It also comprises a thermal management system built, including those of coolant, refrigerant, and ram air circuits, along with a controlling algorithm emulating real-world operating conditions.

1D Battery Electric Truck model main parts are battery, e-motor, transmission, powertrain control algorithm, cooling system, HVAC system, and vehicle components.

2.1. Model architecture and components

The one dimensional vehicle model is designed to simulate the energy flow and performance of the heavy-duty BEV truck. The model architecture, as illustrated in Fig. 1, comprises two main electrical circuits: a high voltage (HV) powertrain for propulsion and a low voltage (LV) circuit for auxiliary systems. This integrated approach allows for a comprehensive analysis of the total energy consumption under various operating conditions.

High-Voltage (HV) Powertrain: The core of the HV system is the main traction Battery, representing the 300 kWh energy storage system that supplies high-voltage DC power. This power is channeled to the Inverter_Motor block, which is governed by a powertrain control algorithm. This algorithm interprets the demands of the driving cycle, such as target speed and road gradient, to determine the precise power required. In response to this command, the inverter converts the battery's DC power to AC to drive the motor, which in turn generates the mechanical torque required for motion. During deceleration, this process is reversed for regenerative braking, converting kinetic energy back into electrical energy to recharge the battery. The mechanical output from the motor is fed into the Vehicle block. This block models the physical dynamics of the truck, incorporating key parameters such as vehicle mass, aerodynamic drag, and rolling resistance, to calculate the required tractive force for a given drive cycle.

Table 1
Vehicle specification.

E-truck specification (Unit)	Parameter	Value
Vehicle mass (tons)	m	16
Range (km)	–	200
Tire radius (m)	r	0.447
Rolling resistance coefficient (-)	f_R	0.005
Drag coefficient x Vehicle frontal area (-)	c_W	6.565
Air density (kg/m^3)	ρ	1.2
Gravitational acceleration (-)	g	–
Gradient angle (-)	α	Various
Battery total capacity (kWh)	–	300
Battery initial SOC (%)	–	90
E-motor maximum power (kW)	–	210

Low-Voltage (LV) Auxiliary System: The LV circuit is powered via a DC–DC converter. This critical component steps down the high voltage from the main traction battery to a lower voltage (e.g., 12V) to supply the vehicle's auxiliary systems. This low-voltage power maintains the charge of the 12 V Battery, which acts as an energy buffer and powers the Auxloads. Crucially for this study, the Aux block represents the combined electrical consumption of all non-propulsion components, including the thermal management systems under investigation: the HVAC system (compressor, cabin blower), coolant pumps, and radiator fans.

By simulating the dynamic interactions between these components over a complete driving cycle, the model accurately calculates the total energy drawn from the main battery. This enables a detailed breakdown of energy consumption, distinguishing between the energy used for propulsion and the energy consumed by auxiliary systems, which is fundamental to evaluating the efficiency impact of the PTC heater versus the heat pump system.

3. E-truck specifications & heat pump modeling

3.1. E-truck specification

Table 1 lists the specifications of the e-truck used in the one-dimensional model simulation. Dual electronic motors are employed in this study to achieve the maximum target grade within continuous operation limits. Identical e-motors are selected, and their torque limits are considered to continue your operation. Peak torque limits are time-restricted to protect the motors. Since the cooling system was not finalized in the early stages of the project, peak torque operating conditions were not examined.

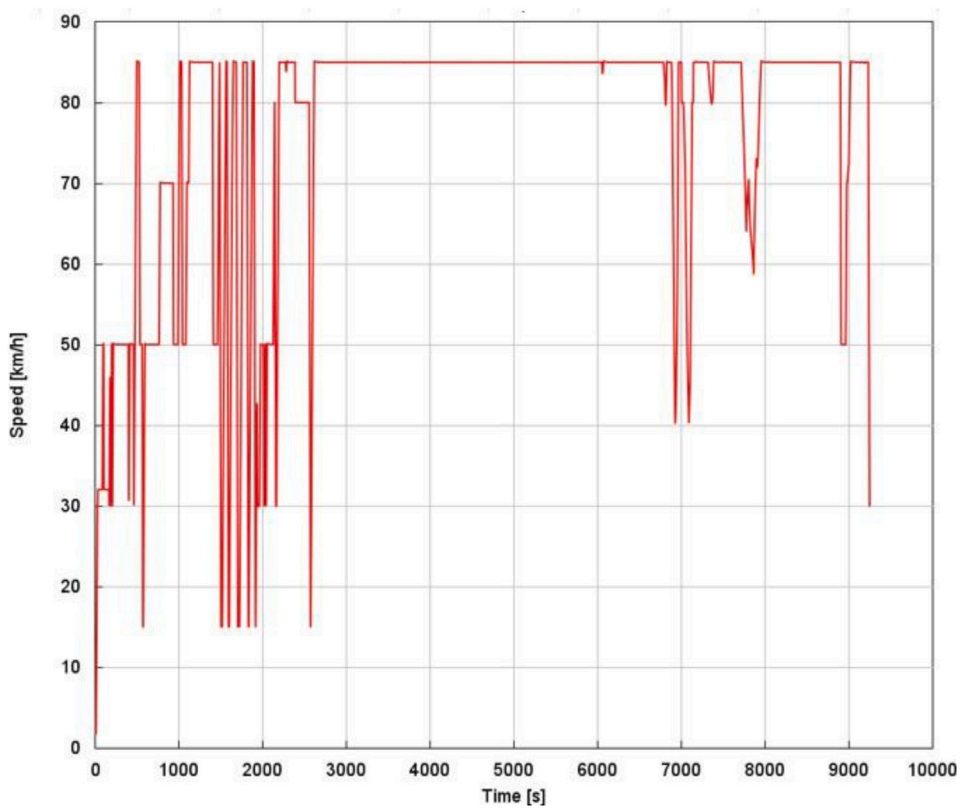

Fig. 2. ORFE cycle.

A 300 kWh battery capacity, a proven success from another BEV Road Truck project, is chosen for the 200 km target range. Energy consumption figures are calculated based on the consumed energy and the difference between the initial SOC and the final SOC at the end of the cycle.

3.2. Driving cycle

On-Road Fuel Economy (ORFE) Cycle is developed to test a vehicle's energy consumption. The central idea under this would be exhaustive analysis of vehicle performance upon varying drive scenarios while measuring energy efficiency. This would entail a wide variety of driving conditions using different speeds and loads, ensuring that approach is both holistic and complete enough for making manufacturers as well as regulatory bodies assess and better understand environment impact and energy use of these vehicles.

In a ORFE cycle analysis, the following assessments are conducted:

•Speed and Load Conditions: Assessing the performance of the vehicle under different speeds and load.

•Energy Consumption: Measurement and evaluation of an energy usage of vehicles.

•Emission Performance: Determining and analyzing vehicle emissions.

•Real-World Simulation: Simulates actual driving conditions to ensure accurate and reliable results.

It is illustrative of the ORFE Cycle, as in Fig. 2 which would yield enhanced energy efficiency Improvements for vehicles and emission reductions. Primarily, it performs for larger engines and commercial applications which usually consume much more energy and produce higher emissions. The ORFE Cycle is a major component in facilitating improvements in vehicles' performance while reducing their environmental impacts within the automotive and environment sectors.

3.3. Heat pump architecture

Heat pumps are systems that revolve around the HVAC system being used for heating the cabin when there is cold, enhancing the efficiency and comfort of a modern EV. Its operation may be simplified into four steps:

1. Heat Absorption: The refrigerant in the evaporator absorbs heat either from the outside air or the battery, thus evaporating into a gas.
2. Compression: The gas is then compressed by the compressor, which raises its temperature.
3. Heat Release: The hot gas moves to the condenser, where it gives up heat to the cabin or battery, condensing back into a liquid.
4. Expansion: The liquid refrigerant goes through the expansion valve, which further drops the pressure and temperature. This process continues in a loop.

Key components of a heat pump system:

1. Compressor: It compresses the refrigerant, increasing its temperature and pressure.
2. Condenser: The hot refrigerant releases its heat to the cabin or battery, cooling down and condensing back into a liquid.
3. Expansion Valve: The expansion valve reduces the pressure of the refrigerant, cooling it down before it
4. Evaporator: The refrigerant absorbs heat from outside air or from the battery and becomes a gas.

3.4. Heat pump component modeling

To ensure the fidelity of the thermal system simulation, each key component of the heat pump circuit was modeled with specific physical and empirical details. This approach enhances the credibility of

the model by grounding it in established engineering practices and component characteristics.

- Refrigerant Fluid: The working fluid selected for the heat pump cycle is R134a (Tetrafluoroethane). The thermodynamic property data for R134a were obtained from the GT-Suite property database, which is based on REFPROP correlations [21]. Although newer alternatives exist, R-134a was chosen for this study due to its extensive use in the automotive industry over the past decades and the availability of well documented, reliable thermodynamic property data, which is crucial for simulation accuracy. Nevertheless, it is important to note that global regulations, such as the EU F-Gas Regulation and the Kigali Amendment to the Montreal Protocol, are progressively restricting the use of high-GWP refrigerants. Consequently, next-generation refrigerants such as R1234yf (with ultra-low GWP) and natural alternatives like R744 (CO_2) are becoming more common. Switching to these refrigerants is expected to change the system's performance: R1234yf offers similar efficiency with reduced environmental impact, whereas R744 can provide improved heating performance at moderate ambient temperatures but requires significantly higher operating pressures and careful frost management strategies.
- Heat Exchangers: The heat exchangers in the model are defined as crossflow flat tube heat exchangers, which represent the typical configuration used in automotive applications. This design was selected because it offers a favorable balance between thermal efficiency, compactness, and acceptable pressure drop. Although counterflow arrangements can theoretically yield higher heat transfer performance, crossflow configurations are widely adopted in vehicle HVAC systems due to packaging and integration constraints. Fig. 3 illustrates the geometric and material property definitions used in the 1D simulation environment, representing a crossflow flat-tube structure. Thermal performance is calculated using the effectiveness-NTU (ϵ-NTU) method, which is directly informed by specific geometric parameters. The frontal area (A_{fr}), which is critical for calculating the air side mass flow rate, is determined by the core height ($H = 225$ mm) and width ($W = 180.7$ mm) as shown in Eq. (1).

$$A_{fr} = H \times W \tag{1}$$

  Other key dimensions defined in the model include a core depth ($D = 42$ mm) and inlet/outlet connection diameters ($D_{m1}, D_{m2} = 13.5$ mm), which are essential for accurate fluid dynamic and pressure drop calculations.
- Compressor efficiency: The performance of the electric Air Conditioning (A/C) compressor is defined using empirical efficiency maps derived from manufacturer data. These maps correlate the compressor's volumetric and isentropic efficiency with its primary operating conditions, namely the compressor speed and the pressure ratio across the component, to accurately predict the refrigerant mass flow rate and the required power consumption.

The heat transfer coefficients within the heat exchangers (HX) are calculated using the Nusselt (Nu) number correlation approach, which is a standard feature of GT-SUITE. According to the methodology, the solver employs a generalized Nusselt correlation expressed as:

$$Nu = C \cdot Re^m \cdot Pr^{1/3} \tag{2}$$

where Re and Pr denote the Reynolds and Prandtl numbers, respectively. The correlation coefficients C and the exponent m are not assumed a priori but instead are derived from empirical data. In the present study, this is implemented through the use of an HxNuMap object, in which measured heat transfer performance data (e.g., temperatures and flow rates) are supplied. During a preprocessing step, the solver fits the experimental data to the correlation, thereby determining the specific values of C and m for three distinct flow regimes: laminar, transition, and turbulent.

3.5. System architecture and cooling mode operation

The thermal management system of the BEV truck is based on a versatile heat pump architecture, capable of providing both cabin and battery cooling. The detailed schematic of the refrigerant circuit operating in cooling mode is presented in Fig. 4. The operation follows a standard vapor-compression refrigeration cycle, with specific pathways controlled by a series of shut-off valves to direct cooling where it is needed.

The cooling cycle can be described in four primary stages:

1. Compression: The cycle initiates at the A/C Compressor, which draws low-pressure refrigerant vapor from the evaporator circuit. It compresses this vapor into a high-pressure, high-temperature superheated gas, providing the energy required to drive the cycle.
2. Condensation: The high-pressure gas is directed to the front-mounted heat exchangers, namely the condenser and the primary radiator. In this stage, heat is rejected from the refrigerant to the ambient air, assisted by the cooling fan. As the refrigerant releases heat, it condenses into a high-pressure, subcooled liquid. The liquid then passes through the receiver dryer unit, which removes moisture and impurities from the system.
3. Expansion: The high-pressure liquid refrigerant flows toward the expansion devices. The system is designed with two parallel cooling circuits: one dedicated to the cabin evaporator and another to the battery chiller. In the operating mode described, the shut-off valves direct refrigerant to the cabin cooling circuit. The refrigerant passes through a thermostatic expansion valve (TXV), where it undergoes a rapid pressure drop (throttling process). This results in a low-pressure, low temperature two-phase mixture. The TXV regulates the refrigerant mass flow based on feedback from a dedicated sensor, ensuring optimal cooling performance and preventing liquid refrigerant from returning to the compressor.
4. Evaporation: The low temperature refrigerant mixture enters the cabin evaporator, located within the HVAC unit. The cabin blower forces warm air over the evaporator fins, allowing the refrigerant to absorb heat from the air. This causes the liquid portion of the refrigerant to boil and transition into a low-pressure vapor. As a result, the air is cooled and recirculated into the cabin. The refrigerant vapor exiting the evaporator then returns to the inlet of the compressor, completing the cycle.

This architecture provides the flexibility to cool the cabin and battery independently or simultaneously by actuating the relevant shut-off valves, thereby managing the key thermal loads within the electric vehicle.

To provide deeper thermodynamic insight, the heat pump is modeled based on the First Law of Thermodynamics. The system is analyzed as a vapor-compression cycle operating in heating mode. The primary states of the refrigerant are defined as: State 1 (compressor inlet, saturated vapor), State 2 (compressor outlet, superheated vapor), State 3 (condenser outlet, saturated liquid), and State 4 (expansion valve outlet, two-phase mixture).

The energy balance for each key component is formulated as follows:

Compressor. The power consumed by the compressor is determined by the change in enthalpy of the refrigerant, where $\dot{m}_{ref}$ is the refrigerant mass flow rate:

$$\dot{W}_{comp} = \dot{m}_{ref}\,(h_2 - h_1) \tag{3}$$

Condenser. The useful heat delivered to the cabin ($\dot{Q}_{cabin}$) is the heat rejected by the refrigerant in the condenser:

$$\dot{Q}_{cabin} = \dot{m}_{ref}\,(h_2 - h_3) \tag{4}$$

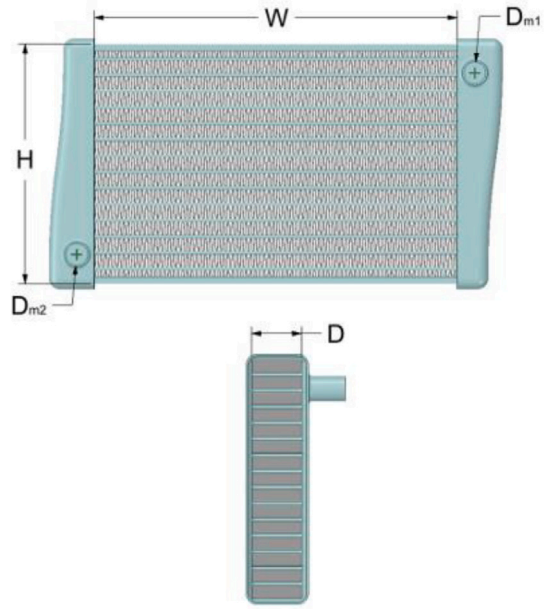

Fig. 3. Geometric and material property definition for a typical flat-tube heat exchanger in the 1D simulation environment.

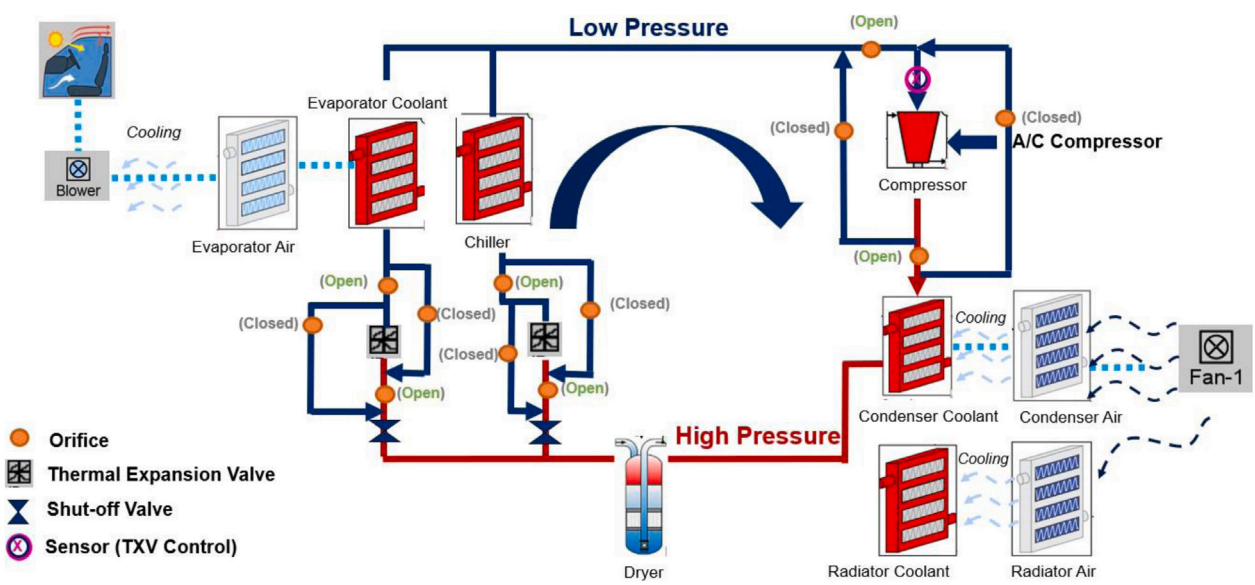

Fig. 4. Heat-pump cooling mode.

Expansion valve (TXV)**.** The throttling process in the expansion valve is assumed to be isenthalpic, meaning that the enthalpy remains constant:

$$h_3 = h_4 \tag{5}$$

Evaporator. The heat absorbed from the ambient air ($\dot{Q}_{evap}$) by the refrigerant in the evaporator is given by:

$$\dot{Q}_{evap} = \dot{m}_{ref}\,(h_1 - h_4) \tag{6}$$

These equations form the basis for evaluating the system's performance from a thermodynamic perspective, going beyond simple energy consumption metrics.

3.6. Heat pump operation in heating mode

To provide cabin heating efficiently, the system reconfigures the refrigerant circuit to operate as a heat pump, effectively reversing the flow of thermal energy compared to the cooling mode. The architecture for this mode is illustrated in Fig. 5. The key principle is to absorb low-grade heat from the cold ambient air and "pump" it into the cabin.

The heating cycle leverages the same core components but alters their roles through valve actuation:

1. Compression**:** As in the cooling cycle, the A/C Compressor takes in low-pressure refrigerant vapor and compresses it into a high-pressure, high-temperature gas.
2. Indoor Condensation**:** In heating mode, the shut-off valves direct the high-pressure gas away from the external heat exchangers and toward the indoor HVAC unit. The refrigerant flows into the cabin evaporator, which in this mode operates as an indoor condenser. The cabin blower forces cold cabin air across its fins, allowing the refrigerant to reject heat and condense into a high pressure liquid. The rejected heat warms the cabin air, which

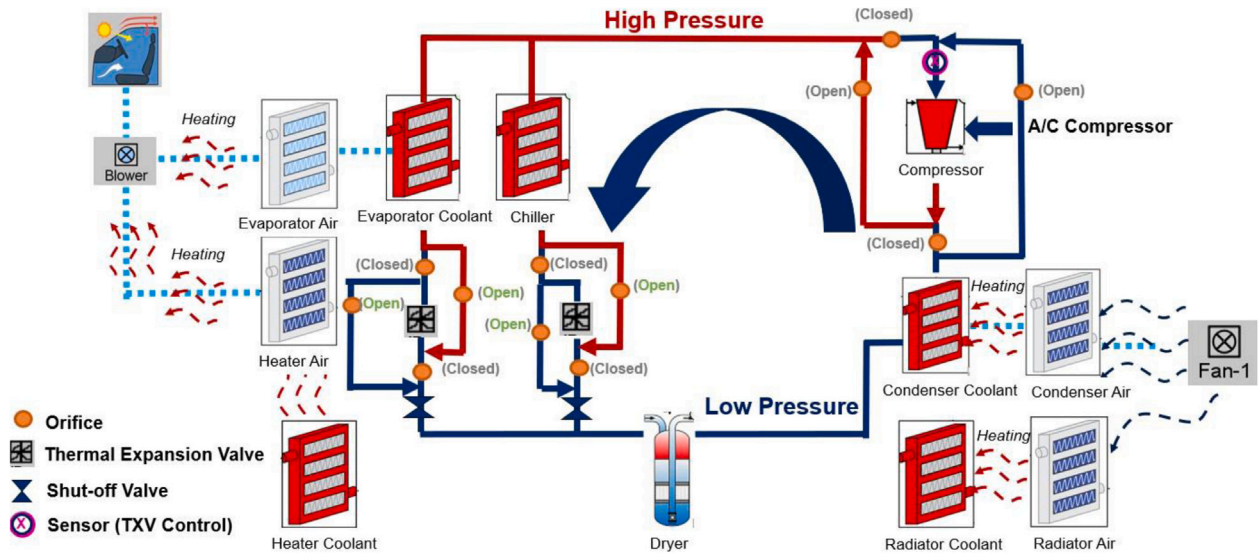

Fig. 5. Heat-pump heating mode.

is then circulated inside the vehicle, as represented by the red dashed arrows.

3. Expansion: After exiting the indoor condenser, the high-pressure liquid refrigerant is routed to the expansion valves located upstream of the outdoor heat exchangers. Here, the refrigerant undergoes a throttling process, leading to a substantial reduction in pressure and temperature, and transforming into a cold, low-pressure two-phase mixture.
4. Outdoor Evaporation: The cold refrigerant mixture then enters the external heat exchangers, which in this mode function as evaporators. Assisted by the cooling fan, ambient air is blown across their fins. Since the refrigerant is colder than the ambient air, it absorbs thermal energy from the environment. This energy input causes the refrigerant to boil and fully evaporate into a low pressure gas, which then returns to the compressor inlet, thereby completing the cycle.

This configuration allows the system to transfer low-grade thermal energy from the ambient environment into the cabin, achieving a Coefficient of Performance (COP) significantly greater than unity, which is impossible for a direct electric heater like a PTC. The system's ability to utilize the Chiller circuit in a similar manner also provides a mechanism for heating the battery pack in cold conditions.

3.7. Model inputs

The heat pump system model takes into account the following parameters as input parameters:

- Vehicle speed — The speed of the vehicle affects the heat exchange dynamics.
- Fan speed — Fan speed affects the airflow and heat dissipation.
- A/C button status — The AC button can be either in "on" or "off" state-value.
- Blower speed/status — Operative status and speed of the blower unit.
- Ambient temperature — Outside temperature affecting heat transfer.
- Cabin initial temperature — Temperature at the beginning in the vehicle cabin.
- Battery initial temperature — An initial temperature for the vehicle battery.
- Solar Flux — Radiative energy from the sun (zero for this model).

Table 2
Model inputs and outputs of the heat pump system.

Model inputs	Model outputs
Vehicle speed	Compressor speed
Fan speed	Compressor outlet pressure
AC button status	Evaporator sensor temperature
Blower speed/status	Cabin temperature
Ambient temperature	Cabin/Evap mass flowrate
Cabin initial temperature	Condenser outlet temperature
Battery initial temperature	TXV mass flowrate
Solar Flux (=0)	Battery temperature

3.8. Model outputs

The outputs of the heat pump system model are mentioned here after:

- Compressor speed — The speed at which the compressor is running.
- Compressor outlet (ref) pressure — The pressure in the refrigerant fluid found at the outlet of the compressor.
- Evaporator sensor (air) temperature — The temperature at the evaporator sensor is determined.
- Cabin (air) temperature — The temperature of the air inside the vehicle cabin.
- Cabin/Evap (air) mass flowrate — Air mass flow rate through the cabin and evaporator assembly.
- Condenser (ref) outlet temperature — Refrigerant temperature at the outlet of the condenser.
- TXV (ref) mass flowrate — Mass flow rate of refrigerant going through the thermal expansion valve.
- Battery temperature — The temperature of the vehicle battery during operation.

Table 2 lists the inputs and outputs of thermal system used in the one-dimensional model simulation.

4. Methodology

It lays down a systematic methodology for checking the energy consumption as well as cabin performance of an electric vehicle under distinct heating scenarios. The overall methodology consists of following meaningful key steps:

1. Developing a 1D Model :
   In this work, a 1D thermal and energy management modeling has been carried out for simulating energy and thermal behavior itself in the life of a vehicle. The model integrates the important subsystems comprising battery, electric motor, and HVAC systems.
2. Definition of Inputs and Requirements :
   The operational requirements of this particular system were driven directly from testing conditions similar to those during actual driving, along with real-world environmental influences. The ambient temperature, the speed of cars, and thermal loads were identified as the necessary input parameters forming a basis for making realistic simulation framework.
3. Set Control Algorithms : There have been established control algorithms for battery and electric motor to control the energy distribution effectively with performance edge for the vehicles in their heat stability.
4. Integration of Cooling System Data :
   This provided compatibility to the simulation environment by converting this analysis into 1D representation, thus enhancing the model's accuracy from collecting 3D CAD data from design team of the cooling system components.
5. Simulation and Performance Analysis :
   The final version of the 1D model was utilized in developing a simulation model of varying operational scenarios. An energy consumption and thermal performance analysis of the cabin had been conducted under varying operational scenarios to evaluate the efficiency of the heat pump and the PTC heater systems.

4.1. Heating & cooling methodology

Several factors must be considered when determining the thermal requirements for heavy duty vehicles, such as environmental conditions, battery type, capacity, and the vehicle's usage profile.

Standard heating technologies used in heavy-duty vehicles to heat the cabin include resistive heating, which uses a PTC heater, and heat pump heating. These are also controlled with thermal management systems.

In general, the PTC heater consumes more energy than a heat pump but can supply a stable amount of heat even under low-temperature conditions [24]. In EVs, when the PTC heating element is to heat cabin air, all power is drawn from the battery power source [25]. Main heat pump systems use a compressor, condenser, evaporator, and expansion valve to provide heating and cooling functions. Compressor performance depends on pressure ratio, compressor speed and relative displacement [26].

Thermal management systems monitor the temperature of batteries and cabin to ensure that they stay within optimal temperature to enhance performance and longevity by preventing overheating or becoming too cold [27–29].

Two TMS control algorithms were established to compare vehicle models with cabin heating using PTC heater and heat pump system. The focus of our study was to evaluate the impact of systems on the energy efficiency of the vehicle, a topic of great interest in the field of automotive technology and thermal management systems [30].

In the model with cabin heating using PTC heater, the following components work differently:

1. AC Compressor: The AC compressor cools the batteries through chiller and/or cabin through evaporator depending on the TMS control state.
2. PTC Heater**:** The PTC heater heats the batteries and/or cabin depending on the TMS control state.

In the model with cabin heating using heat pump, control algorithm components work as described below:

1. AC Compressor**:** Unlike the first model, the second model has a specific function where the AC compressor heats the cabin using the heat pump. The AC compressor also continues its primary function of cooling the batteries and/or cabin and, depending on the TMS control state.
2. PTC Heater: The PTC Heater in the second model only heats the batteries depending on the TMS control state.

Conditions for battery heating to be activated, expressed in Eq 1. If the battery coolant temperature is below 40 °C and the battery temperature is below 20 °C, battery heating is turned ON.

$$\text{Battery Heating} = \begin{cases} \text{ON} & \text{if } T_{\text{coolant}} < 40°\text{C and } T_{\text{battery}} < 20°\text{C} \\ \text{OFF} & \text{otherwise} \end{cases} \tag{7}$$

Conditions for battery cooling to be activated, expressed in Eq 2. If battery temperature is above the 27 °C, battery cooling is ON.

$$\text{Battery Cooling} = \begin{cases} \text{ON} & \text{if } T_{\text{battery}} > 27°\text{C} \\ \text{OFF} & \text{otherwise} \end{cases} \tag{8}$$

Conditions for cabin heating to be activated, expressed in Eq 3. If cabin temperature is below 22 °C, then cabin heating is ON. If cabin temperature is above 24 °C, then cabin heating is OFF.

$$\text{Cabin Heating} = \begin{cases} \text{ON} & \text{if } T_{\text{cabin}} < 22°\text{C} \\ \text{OFF} & \text{if } T_{\text{cabin}} > 24°\text{C} \end{cases} \tag{9}$$

Conditions for cabin cooling to be activated, expressed in Eq 4. If cabin temperature is below 24 °C, then cabin cooling is OFF. If cabin temperature is above 26 °C, then cabin cooling is ON.

$$\text{Cabin Cooling} = \begin{cases} \text{ON} & \text{if } T_{\text{cabin}} > 26°\text{C} \\ \text{OFF} & \text{if } T_{\text{cabin}} < 24°\text{C} \end{cases} \tag{10}$$

4.2. Thermal system energy consumption methodology

The over-time energy consumption values for both the PTC and Air Compressor were extracted to analyze the energy data for high-voltage and low-voltage components. The total HV auxiliary consumption was calculated by summing up the energy consumed by the PTC and A/C Compressor over the entire cycle.

$$\text{Total HV Aux Consumption} = \text{PTC} + \text{A/C Compressor} \tag{11}$$

The exact process was followed for the LV components to find the total LV energy consumption. The Total LV Aux Consumption was found, and the energy consumption values for the Battery Pump, HT Pump, HVAC Cabin Blower, and 4x Fan over time were extracted.

$$\text{Total LV Aux Consumption} = \text{Pumps} + \text{Cabin Blower} + 4 \times \text{Fan} \tag{12}$$

For each component, the energy consumption values over the entire cycle were summed up to determine the total energy consumed by that component during the cycle. Then, the total HV energy consumption was calculated by summing up the energy consumption values of the HV components.

$$\text{Total Aux Consumption} = \text{Total HV} + \text{LV Consumption} \tag{13}$$

The calculated energy consumption per 100 km for the current model was compared with the baseline model to identify any differences or improvements. When calculating the percentage improvement, the percentage difference between the total auxiliary consumptions was calculated, and this value was multiplied by 100 to determine the improvement as a percentage.

The percentage improvement in battery energy consumption was determined by dividing the difference in total auxiliary consumptions by the battery's usable capacity and multiplying by 100.

$$\text{Battery Usable Capacity} = \text{Usable SOC} \times \text{Battery Capacity} \tag{14}$$

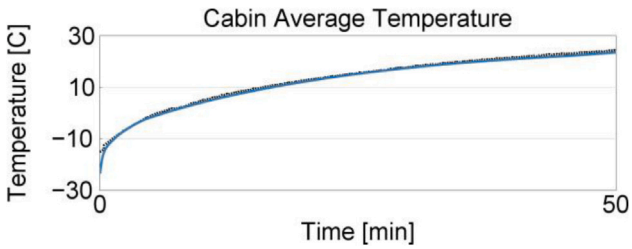

Fig. 6. Comparison of simulated cabin temperature with experimental test data at −18 °C ambient temperature (Dashed line: Experimental results; Colored line: Model predictions).

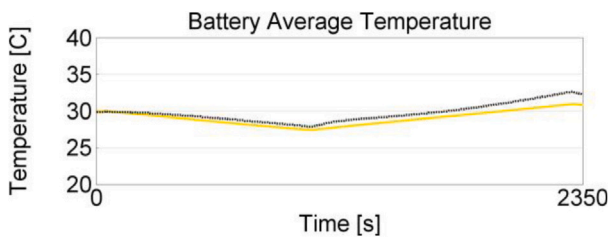

Fig. 7. Comparison of simulated vehicle performance with experimental vehicle test data. (Dashed line: Experimental results; Solid line: Model predictions).

4.3. Model validation

To ensure the reliability and predictive accuracy of the developed 1D thermal model, a crucial validation step was undertaken by comparing its predictions against experimental data from a dedicated HVAC cabin test. The measurement data used for model validation were obtained from vehicle level HVAC tests performed according to an internal procedure in a controlled environment. During the tests, the vehicle was driven under predefined operating conditions while the HVAC system was sequentially operated in different modes (A/C ON, MAX A/C ON, and return to A/C ON), each maintained for approximately 10 min. This setup ensured repeatability and allowed consistent evaluation of the system's thermal response. The profile consists of four distinct phases: (i) a 10 min cold idle phase, (ii) a 20 min steady driving phase, (iii) a 15 min high-speed driving phase or the vehicle's maximum speed, and (iv) a 10 min warm idle phase. This sequence ensures that the HVAC system is evaluated under representative cold start, cruising, high load, and idle conditions. The test was conducted at an ambient temperature of −18 °C. As depicted in Fig. 6, the model's predicted cabin temperature (solid line) shows strong agreement with the experimental measurements (dashed line). The simulation results demonstrate that the model is capable of predicting the cabin temperature within an acceptable deviation of ±3 °C relative to the real-world test data. This close correlation confirms the model's capability to accurately capture the transient thermal behavior of the vehicle cabin under cold ambient conditions, thereby enhancing the confidence in the simulation-based findings presented in this study.

Furthermore, the thermal model's accuracy for battery behavior was validated through a comparison with real-world vehicle test data for average battery temperature. As illustrated in Fig. 7, the model's predictions for the average battery temperature (solid line) closely match the experimental measurements (dashed line). The model consistently estimates the average battery temperature with an accuracy of ±1.5 °C. This rigorous validation process, encompassing both cabin and battery thermal dynamics, significantly reinforces the credibility and robustness of the simulation results, confirming the model's suitability for analyzing the thermal performance of BEV trucks under various operating conditions.

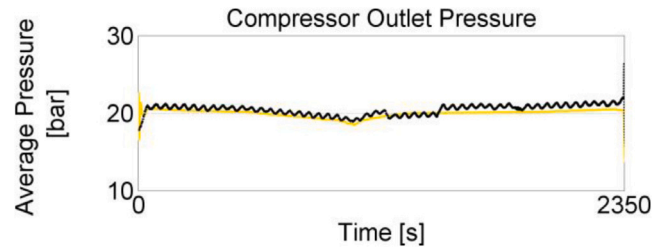

Fig. 8. Comparison of simulated compressor outlet pressure with experimental test data. (Dashed line: Experimental results; Colored line: Model predictions).

Finally, the model's ability to accurately predict the compressor's performance, a key component of the heat pump system, was verified by comparing simulated compressor outlet pressure with experimental test data. As shown in Fig. 8, the model's predictions (colored line) exhibit a strong agreement with the measured values (dashed line) throughout the test duration. This consistency in predicting a critical operational parameter further strengthens the validity of the overall thermal management system model, confirming its capability to replicate real-world operating conditions of the heat pump system.

Since no experimental test campaign with a vehicle equipped with a heat pump was available at the time of this study, the validation was performed using measurements from the baseline PTC heater configuration. Therefore, the validation of the model focused on the comparison of measured and simulated temperature and pressure profiles, which provide reliable indicators of transient thermal behavior of the system.

5. Results & discussion

This section aims to quantify the impact of substituting a heat pump for a PTC heater on a vehicle's range, energy consumption, and overall thermal efficiency.

5.1. Vehicle energy consumption & range improvement results

Heat pumps in Battery Electric Vehicles (BEVs) can significantly impact energy consumption, particularly in cold climates. Heat pumps use less energy than classic resistance heating systems, which increases the vehicle's range.

Control strategies, such as those mentioned in Section 2, should be developed to improve the energy efficiency of heat pump systems. These solutions aim to reduce wasteful energy usage. For example, altering the heat pump's working mode dependent on the outside temperature might increase energy efficiency.

The 1D model that is developed calculates the energy consumption of components within the thermal system. This includes the PTC heater and AC compressor in the high-temperature (HT) circuit and pumps, blowers, and radiator fans in the low-temperature (LT) circuit.

Energy management system improvement is aimed by calculating and comparing the auxiliary consumptions in our vehicle. Table 3 below shows the thermal system consumption of an electric vehicle with different energy management systems. In this study, a total of several models were compared: cabin heating with PTC (0 °C ambient) and cabin heating using a heat pump without PTC connection (0 °C ambient).

The energy consumption values in the table are calculated by measuring the power usage of different vehicle components over a given distance (197 km in this case). These components are divided into high voltage (HV) outputs, low voltage (LV) Thermal Management System (TMS) outputs, and Auxiliary (Aux) outputs.

Figs. 9 and 10 show the percentage distributions in a way that visualizes the comparison of energy consumption for the two heating configurations, thus clearly demonstating teaching of the impact of the two systems on overall vehicle efficiency.

Table 3
Thermal system energy consumption comparison.

Components	Heat pump	PTC heater	Units
PTC	0	5.53	kWh
AC_Compressor	3.51	0	kWh
Total HV Output	3.51	5.53	kWh
Battery_Pump	0.26	0.25	kWh
Cabin_Pump	0.32	0.41	kWh
HT_Pump	0.49	0.49	kWh
HVAC_Cabinblower	1.28	1.34	kWh
Fan 1	0.04	0.19	kWh
Fan 2	0.04	0.19	kWh
Fan 3	0.04	0.21	kWh
Fan 4	0.04	0.19	kWh
Total LV TMS Output	2.5	3.3	kWh
Total Aux Output	6	8.8	kWh
Total Distance	197	197	km
Total Aux Consumption	3.06	4.47	kWh/100km

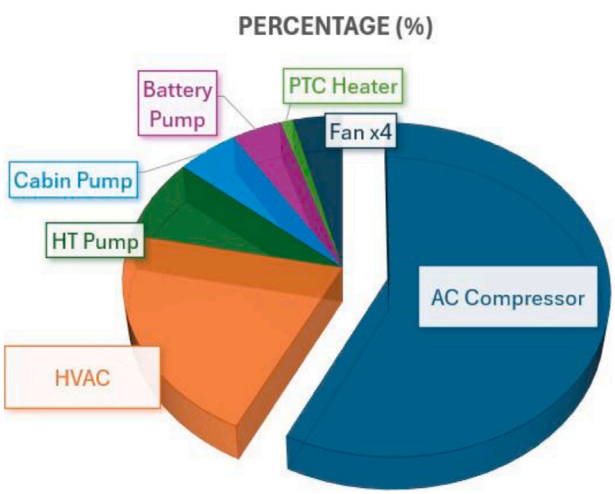

Fig. 9. Heat pump energy consumption distribution.

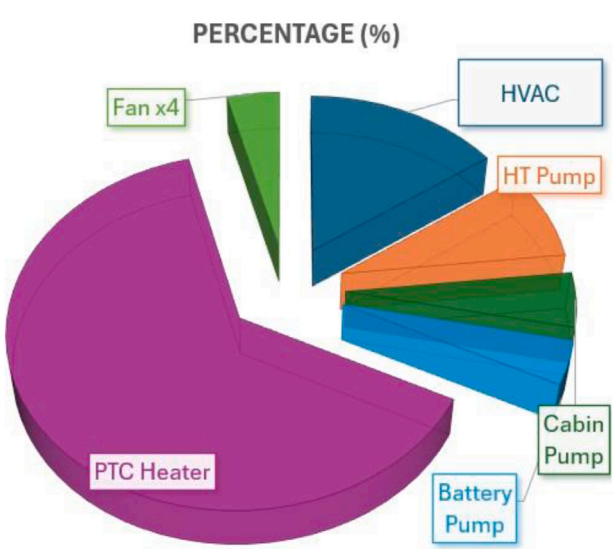

Fig. 10. PTC heater energy consumption distribution.

5.2. Total thermal management system outputs

The consumption of the HV system consists of the PTC heater and the AC compressor. At an ambient temperature of 0 °C, the heat pump has an energy advantage compared to the PTC heater for cabin heating. Heating from outside will help heat pumps work more efficiently. Heating needs are high even in very cold conditions, but the heat pump is a significant improvement over the PTC heater while heating less than 3.51 kWh for heating. The PTC heater may consume up to 5.53 kWh electrical energy making it less efficient, mainly at colder temperatures.

The consumption for the Low Voltage (LV) Thermal Management System (TMS) comprises different components such as the battery pump, cabin pump, HT pump, cabin blower, and fan. The same amount of energy is consumed by the battery pump in both heating systems. For heat pump use, this is 0.26 kWh, and for the PTC heater, it is 0.25 kWh, which is almost equal. Since battery thermal regulation is very important in both configurations, the difference in energy consumption is minimal. However, the cabin pump indicates a more marked difference, as this consumes 0.32 kWh in the heat pump and 0.41 kWh in the PTC heater because there is a need for active heat distribution in the PTC heater increasing the workload for the pump. Fig. 9 shows the energy consumption distribution for the heat pump system showing the share of each auxiliary component in total consumption and Fig. 10 shows the energy consumption distribution for the PTC heater system, highlighting the higher share of fans and the PTC heater compared to the heat pump configuration.

The HT pump operates uniformly consuming 0.49 kWh in the two systems. This is expected because this pump ensures thermal balance in the heating circuit of the vehicle whether it has a PTC heater or a heat pump. Very little increase is seen in the energy consumption of the HVAC cabin blower with the use of the PTC heater from 1.28 kWh in the heat pump system to 1.34 kWh in the PTC heater system. Making strong airflow circulation within will make effective distribution of the heat produced because the PTC heater does not rely on the normal heat transfer process, thus reducing the workload of the blower.

The PTC heater system yields much higher energy consumption from the four cooling fans (Fan 1–Fan 4), with each consuming 0.19–0.21 kWh, against 0.04 kWh in the heat pump system. Additional circulation is required to supply the heat generated by the PTC heater because the heating element relies only on electrical resistance heating, increasing electricity demand in fans used. Therefore, the consumption in this regard will be higher for the entire LV TMS as performed by the PTC heater system than by the heat pump system. Where P_b, P_c, P_h, P_{hvac}, and P_f represent the energy consumption of the battery pump, the cabin pump, the HT pump, the HVAC cabin blower and the Fans.

$$\text{Total LV TMS Output} = \sum(C_b + C_c + C_h + C_{hvac} + C_f) \tag{15}$$

which results in 2.5 kWh for heat pump and 3.3 kWh for PTC heater.

5.3. Auxiliary (aux) consumption and efficiency metrics

- Total Aux Output (6–8.8 kWh): The total auxiliary consumption includes all non-propulsion-related energy use (HVAC, pumps, and fans).
- Total Distance (197 km): This is the driving distance for which the energy consumption values are recorded.

The total auxiliary consumption per 100 km is calculated as:

$$\text{Total Aux Consumption} = \frac{\text{Total Aux Output}}{\text{Total Distance}} \times 100 \tag{16}$$

For Heat Pump:

$$\frac{6}{197} \times 100 = 3.06 \text{ kWh/100 km} \tag{17}$$

For PTC heater:

$$\frac{8.8}{197} \times 100 = 4.47 \text{ kWh/100 km} \tag{18}$$

The numerical improvements are calculated from the component energy consumptions listed in Table 3. Using the values for total

Table 4
Thermal system energy comparison.

Component	HP	PTC	Unit
Thermal auxiliary load improvement	31.64	–	%
Battery energy improvement	1.18	–	%

auxiliary consumption (Heat Pump: $Q_{\text{HP}} = 6.00$ kWh; PTC Heater: $Q_{\text{PTC}} = 8.80$ kWh), the absolute auxiliary energy savings is

$$\Delta Q_{\text{aux}} = Q_{\text{PTC}} - Q_{\text{HP}} = 8.80 - 6.00 = 2.80 \text{ kWh}. \tag{19}$$

The percentage reduction in auxiliary thermal load (see Table 4) is computed as the relative change with respect to the PTC case:

$$\text{Aux. Improvement} = \frac{\Delta Q_{\text{aux}}}{Q_{\text{PTC}}} \times 100 = \frac{2.80}{8.80} \times 100 \approx 31.6\% \tag{20}$$

To convert auxiliary energy savings into an equivalent improvement in battery energy consumption, Eq. (14) in the Methodology is used:

Thus, the percentage improvement in battery energy is

$$\begin{matrix}\text{Battery}\\ \text{Improvement}\end{matrix} = \frac{\Delta Q_{\text{aux}}}{\text{Battery usable capacity}} \times 100. \tag{21}$$

The usable battery capacity is calculated from the nominal cell parameters and the defined state-of-charge (SOC) window. The total ampere-hour capacity is determined by the number of cells in parallel:

$$Q_{\text{total}} = N_{\text{cell}} \times N_{\text{parallel}} \times C_{\text{cell}} = 148 \times 3 \text{ Ah} = 444 \text{ Ah}. \tag{22}$$

The nominal total energy content of the pack is then

$$E_{\text{total}} = \frac{V_{\text{total}} \times Q_{\text{total}}}{1000} \quad [\text{kWh}], \tag{23}$$

where V_{total} is the total pack voltage.

The usable energy is defined by the usable SOC window:

$$E_{\text{usable}} = E_{\text{total}} \times SOC_{\text{usable}}. \tag{24}$$

Imposing the condition that $E_{\text{usable}} = 237$ kWh with $Q_{\text{total}} = 444$ Ah and $SOC_{\text{usable}} = 0.9$, the corresponding total voltage is obtained as

$$V_{\text{total}} = \frac{E_{\text{usable}} \times 1000}{Q_{\text{total}} \times SOC_{\text{usable}}} = \frac{237 \times 1000}{444 \times 0.9} \approx 594 \text{ V}. \tag{25}$$

Therefore, with $Q_{\text{total}} = 444$ Ah, $SOC_{\text{usable}} = 0.9$, and $V_{\text{total}} \approx 594$ V, the usable battery capacity is

$$E_{\text{usable}} = 237 \text{ kWh}.$$

The corresponding improvement in overall battery energy (see Table 4) is calculated as the ratio of the saved auxiliary consumption to the usable battery capacity:

$$\text{Battery improvement} = \frac{\Delta Q_{\text{aux}}}{E_{\text{usable}}} \times 100. \tag{26}$$

Substituting $\Delta Q_{\text{aux}} = 2.80$ kWh and $E_{\text{usable}} = 237$ kWh yields

$$\text{Battery improvement} = \frac{2.80}{237} \times 100 \approx 1.18\%.$$

5.4. Thermodynamic performance analysis

To provide deeper technical insights beyond the expected efficiency gains, a thermodynamic analysis based on the First and Second Laws of Thermodynamics was performed.

5.4.1. First law analysis: Coefficient of performance (COP)

The Coefficient of Performance (COP) is a performance metric derived from the first law of thermodynamics, defined as the ratio of the desired thermal output to the required work input.

For the PTC heater, the process is a direct conversion of electrical work to heat. Therefore, its COP is theoretically limited to 1.

$$COP_{PTC} = \frac{\dot{Q}_{out}}{\dot{W}_{elec}} \approx 1 \tag{27}$$

For the **heat pump**, the COP is calculated using the thermodynamic properties of the refrigerant cycle. It represents the ratio of the heat delivered to the cabin to the power consumed by the compressor.

$$COP_{HP} = \frac{\text{Desired Output}}{\text{Required Input}} = \frac{\dot{Q}_{cabin}}{\dot{W}_{comp}} = \frac{h_2 - h_3}{h_2 - h_1} \tag{28}$$

This formulation demonstrates that COP_{HP} can be significantly greater than 1, as it "pumps" existing heat from the ambient ($\dot{Q}_{evap}$) rather than converting work directly into heat.

5.4.2. Second law analysis: Exergy destruction

Although COP compares performance, it does not reveal the sources of thermodynamic imperfection. Exergy analysis, based on the Second Law, quantifies the destruction of useful work potential (irreversibility) in each component. Total exergy destruction ($\dot{I}_{total}$) indicates how much of the high-quality energy input is wasted. The primary sources of exergy destruction ($\dot{I}$) in the cycle are calculated as follows, where T_0 is the ambient temperature.

Compressor**:** Irreversibility arises from non-isentropic compression (e.g., friction).

$$\dot{I}_{comp} = \dot{m}_{ref} T_0 (s_2 - s_1) \tag{29}$$

Condenser**:** Irreversibility is caused by heat transfer across a finite temperature difference between the refrigerant and the cabin air.

$$\dot{I}_{cond} = \dot{m}_{ref} T_0 \left((s_3 - s_2) + \frac{h_2 - h_3}{T_{cabin}} \right) \tag{30}$$

Expansion Valve**:** The throttling process is highly irreversible and a major source of exergy loss.

$$\dot{I}_{TXV} = \dot{m}_{ref} T_0 (s_4 - s_3) \tag{31}$$

Evaporator**:** Similar to the condenser, irreversibility occurs due to heat transfer across a finite temperature difference.

$$\dot{I}_{evap} = \dot{m}_{ref} T_0 \left((s_1 - s_4) - \frac{h_1 - h_4}{T_{ambient}} \right) \tag{32}$$

By quantifying these irreversibilities, we can create a breakdown of the sources of inefficiency. For instance, a pie chart showing the percentage contribution of each component (Compressor, Condenser, TXV, Evaporator) to the total exergy destruction would provide novel technical value. This analysis reveals that while the heat pump is more efficient than a PTC heater, significant potential for improvement exists, particularly by addressing the large exergy losses in the expansion valve and compressor. This provides a clear direction for future optimization research.

5.5. Percentage improvement in COP

Thus, from the provided COP values of the two systems, the percentage improvement of COP in the PTC heater system over the heat pump system can be computed using the following formula:

$$\text{Percentage Improvement} = \frac{COP_{PTC} - COP_{HP}}{COP_{HP}} \times 100$$

Given:

$$COP_{PTC} = 1.37, \quad COP_{HP} = 1.07$$

The percentage improvement is:

$$\text{Percentage Improvement} = \frac{1.37 - 1.07}{1.07} \times 100 \approx 28.04\%$$

Hence, the PTC Heater system gives a performance improvement of about 28.04% over the Heat Pump system in terms of COP.

As seen in the model results, the heat pump system helps to improve the range in electric vehicles. Unlike low-weight passenger cars or commercial vehicles, its contribution to the range decreases in the heavy duty truck application. Reduction in range improvement percentage caused by the decrease of the power consumption allocated to cabin heating, when the tonnage of the vehicle increases. However, the contribution of the heat pump to the range improvement at low air temperature should be examined further by introducing the waste heat of the e-drive cooling line to the inlet of the AC compressor which improves super heat. Furthermore, the effect of using the heat pump for battery heating can be investigated.

6. Conclusion

The stringent emission regulations are forcing OEMs to develop innovative solutions to increase energy efficiency in zero-emission vehicles. Heat pump technology is one of the promising solutions in battery electric vehicles due to its capability of increasing range through lowered energy consumption.

First of all, a one-dimensional model is created to mimic the application of heavy duty trucks. Then the heat pump-based model is created. These two models are integrated to obtain a complete understanding of energy efficiency improvement secured via the heat pump system. The model is run using customer-based road cycle data. The performance of the vehicle and the heat pump system are limited with the customer & supplier based requirements in order to mimic the realistic conditions successfully. The operating conditions are defined in alignment with calibrational requirements of the system.

Two different sets of simulations are performed. These simulations represented the cabin heating with PTC and the cabin heating via a heat pump only. The detailed breakdown of the energy consumption is prepared in both conditions. It is revealed that; the PTC heater and AC compressor are the highest energy consumer parts in auxiliaries of thermal system. The results also show that it is possible to obtain 31.64% efficiency improvement with respect to the PTC heater. Finally, the technology can ensure a 1.18% range improvement in battery energy consumption.

The goal of this study is to make the existing refrigerant system operate as a heat pump with minimal modifications. Therefore, only three electrically controlled valves were added to the system, including a flow diverter, and the existing TXVs were replaced with EXVs. The evaporator, condenser, and compressor are kept the same. Those changes make the revised system suitable for both the heat pump and the standard cooling modes. Performing further optimization in the thermal system including the increase in evaporator size and/or the addition of a heat condenser and/or increase heat pump efficiency by utilizing HT waste heat will be the subject of the next examinations. The energy consumption improvement opportunities will be the subject of another manuscript.

CRediT authorship contribution statement

Taha Erkin Tunalı: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis. **Elif Gözen:** Investigation, Methodology, Validation, Formal analysis, Visualization, Writing – original draft. **Emre Özgül:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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